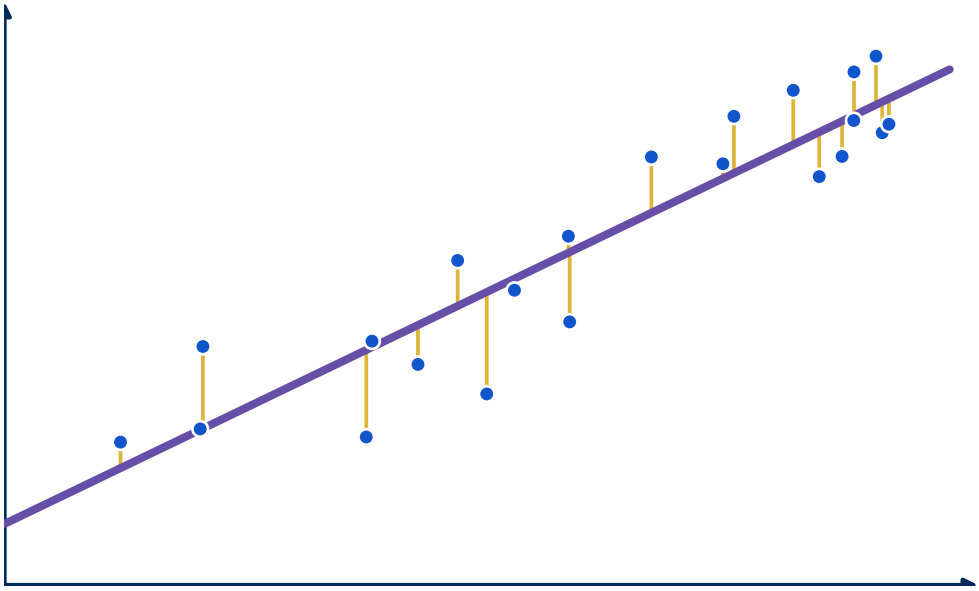


UNIT 01 · CHAPTER 4

Linear Regression

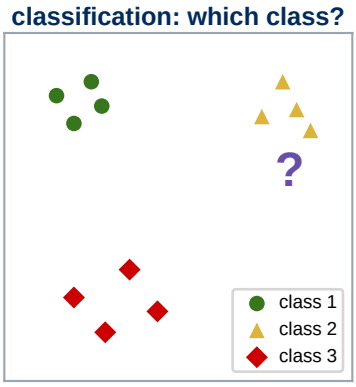
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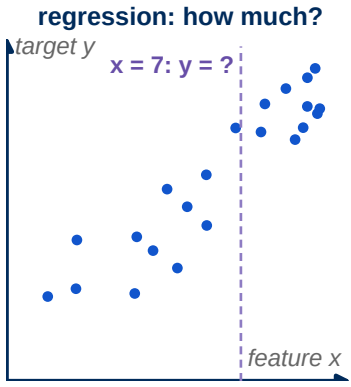
William Theisen · University of Notre Dame

Classification and regression

Our **previous models**, **k-Nearest Neighbors** and **Decision Trees**, were built for **classification**.

A **classification task** is one in which we try to assign a _____ to a sample.





In **regression tasks**, we try and assign a _____ to each sample.

Regression is another form of _____ _____.

Quick check (old exam question): is a regression task supervised or unsupervised learning?

Linear regression

A **linear regression** is a technique to fit a line to our data, allowing us to make predictions.

Regression vocabulary

Unfortunately, because linear regression is so old, it **has a lot of vocab specific to it**:

[4.1] a line with one feature

$$\hat{y} = w_1 x_1 + w_0$$

- $\hat{y}$: _____
- w_1, w_0: _____
- x_1: _____

Matrix notation

We can represent basically everything as **matrices** and **matrix operations**.

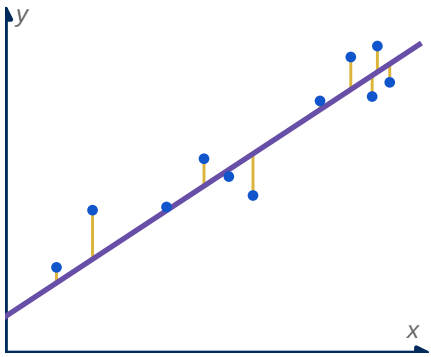
Multiple features

It's rare that we have datasets with only a single feature. The **iris dataset**, a toy dataset, has 4!

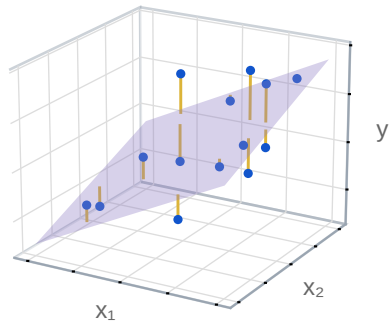
When we perform a **linear regression on** a dataset with **more than one feature**, we call this a _____.

Hyperplanes

1 feature: a line



2 features: a plane

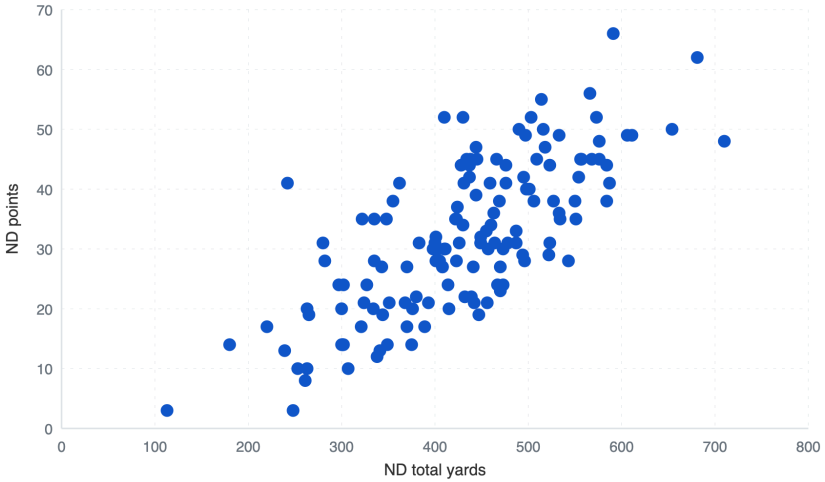


With one feature the model is a line, with two it's a _____ and with more it's a _____.

Quick check (old exam question): what would the equation look like for a multiple linear regression with three features?

Fitting our line

How do we actually find the line that "fits" the best?



Draw the line you think fits best. Which games is it furthest from?

Optimization problems

We need some way to **measure how well our model fits** our data. We can call this a ____ _____.

For **linear regression** we often use **SSE** or ____.

sum of squared errors

$$[4.3] \quad SSE(w) = \sum_i (y_i - \hat{y}_i)^2$$

mean squared error

$$[4.4] \quad MSE(w) = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

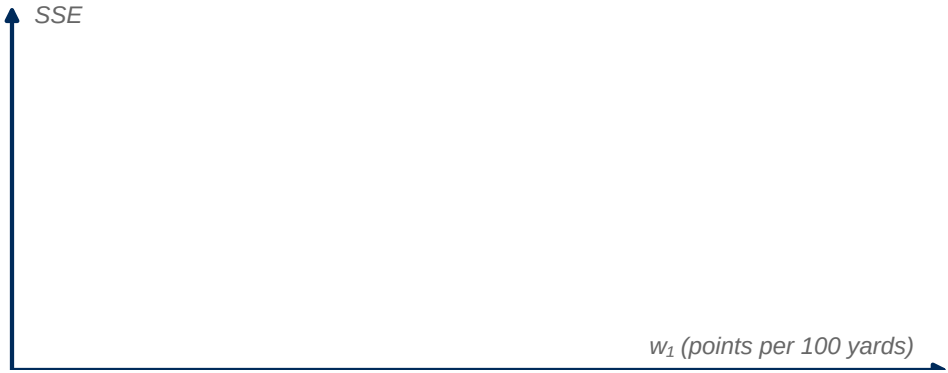
Finding our weights

So given a **loss function**, our **goal** is to _____ it.

When would a **loss function** be **minimized**? When the _____ is ____.

Visualizing loss

We can plot the loss function with respect to **w**:



Ordinary least squares

We can **solve for the gradient equaling 0**:

This is called **Ordinary Least Squares (OLS)**.

Worked example: OLS by hand

game	yards	x_1	y (points)
2019 Michigan	180	2	14
2012 Michigan State	300	3	20
2013 Purdue	400	4	31
2019 Boston College	501	5	40
2017 Temple	606	6	49

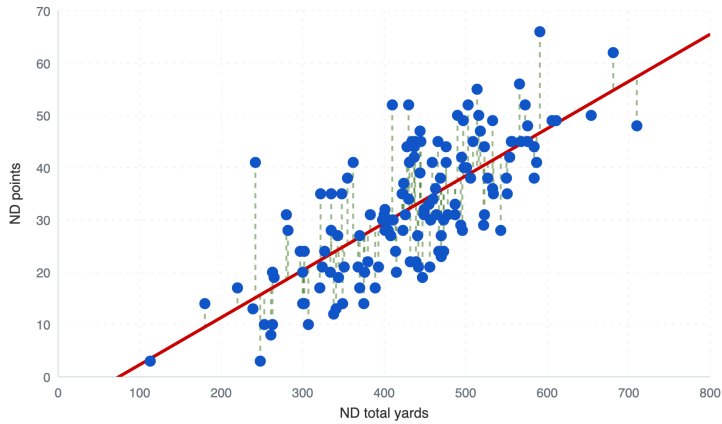
Give X a **column of 1s** for the intercept, then follow the **OLS** recipe.

Reading the weights

So $\hat{y} = 9.0\,x_1 - 5.2$: about _____.

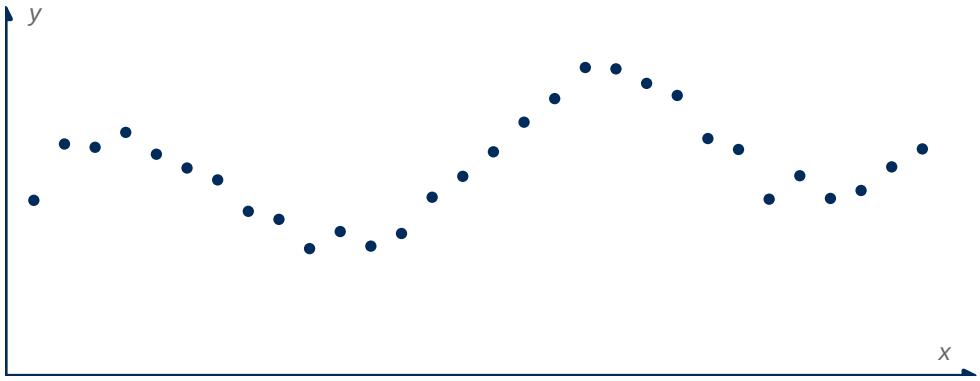
Quick check: in 2018 Notre Dame gained 550 yards against Stanford. What does this line predict, and what is the residual if Notre Dame scored 38?

Linear regression visualization



Representing curves

How could you use a **linear regression** to model a sine wave?



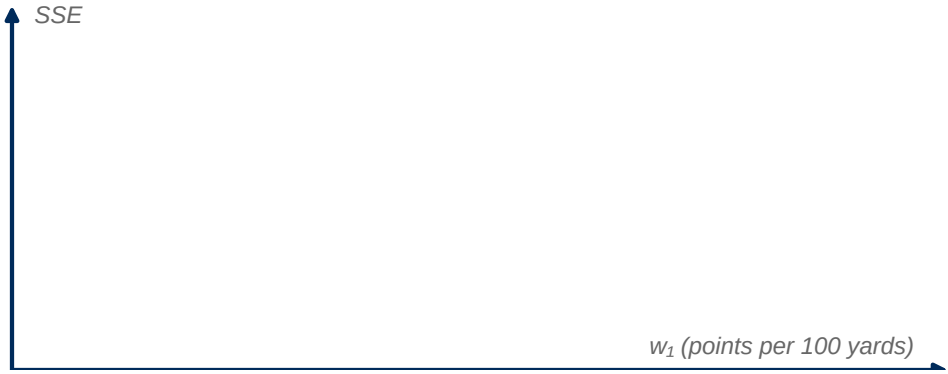
The model only has to be linear in its _____, not in its inputs.

A possible issue

The **weight calculation relies on computing the inverse**, which can be **extremely computationally expensive** and **sometimes just literally impossible**.

An iterative method

Instead of jumping straight to the bottom of the bowl, we can _____ a little at a time.

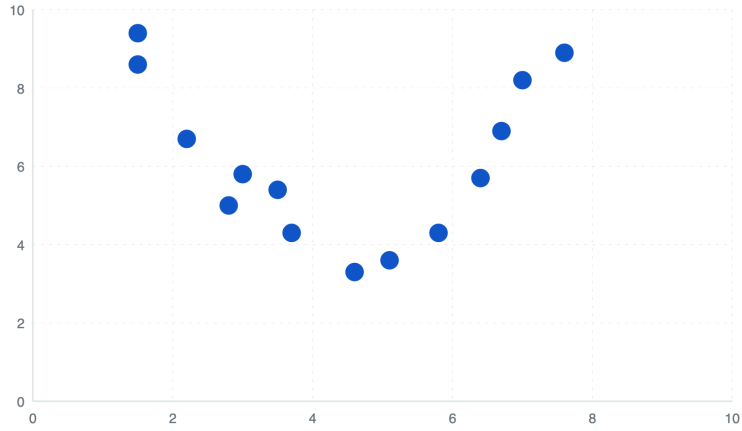


Maximum likelihood estimation

Maximum likelihood estimation (MLE) is a method of estimating the parameters of an _____, **given some observed data**.

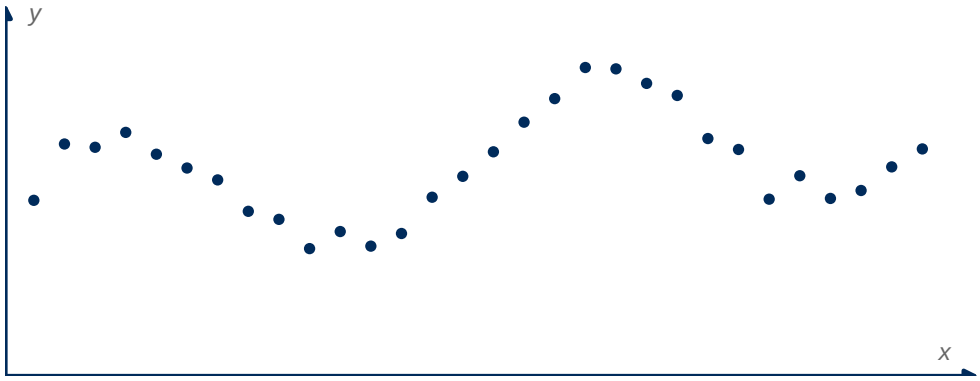
Overfitting

Unfortunately, **Linear Regressions** (a form of MLE), are **prone to overfitting**.

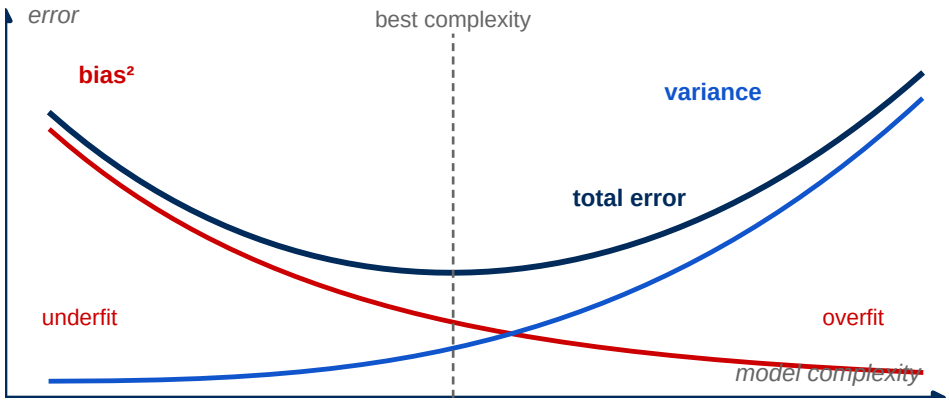


Underfitting

Underfitting is when your model is "too weak" to capture all of the underlying patterns in the data.



Bias and variance



The **bias–variance tradeoff** describes the **relationship between** a **model's complexity**, the **accuracy** of its predictions, and how well it can make **predictions on unseen data** that were not used to train the model.

Practice: an old exam question

x_1	x_2	y
1	1	9
2	2	25
3	3	50
4	4	84
5	5	127

Given the dataset above, answer the following questions **(Old exam question)**:

A. Would a **purely linear model** exhibit **high bias** here or **high variance**? Would it be **overfit** or **underfit**? Explain your reasoning.

B. Design a regression equation that would be able to approximate the dataset.

C. What role does the **SSE** play in **linear regression**? Why does it matter that our resulting parameter space is convex?

D. Look at the x_1 and x_2 columns. What goes wrong if we fit $\hat{y} = w_0 + w_1 x_1 + w_2 x_2$ with **OLS**?

Regularization: L_1 and L_2

To help **prevent overfitting** when using **linear regressions**, we can perform a technique called **regularization**.

The goal of **regularization** is to **encourage models to learn the broader patterns** within the data rather than memorizing it.

L_2 regularization: Ridge

L2 regularization (called **ridge regularization**) **encourages** _____ **weights**.

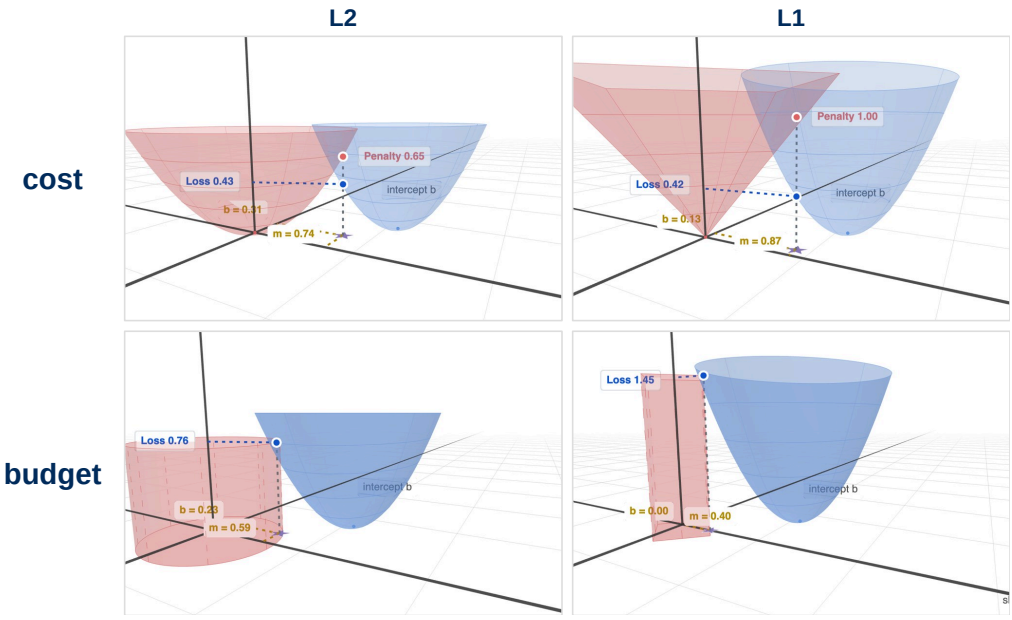
L_1 regularization: LASSO

L1 regularization (also called **LASSO**) leads to _____ and **sparse models**.

Formalizations of regularization

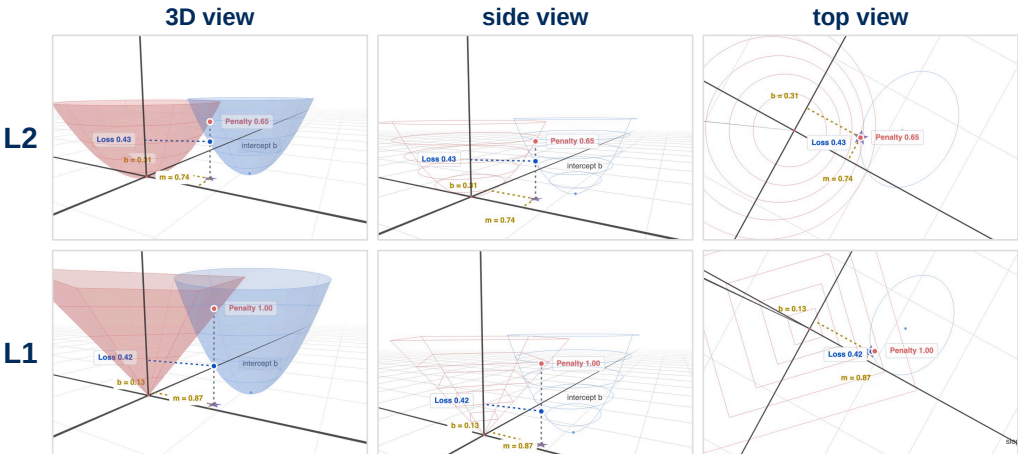
There are two ways to formalize regularization:

- As a _____ placed on the **weights**
- As a _____ for the **weights**



Regularization as a cost

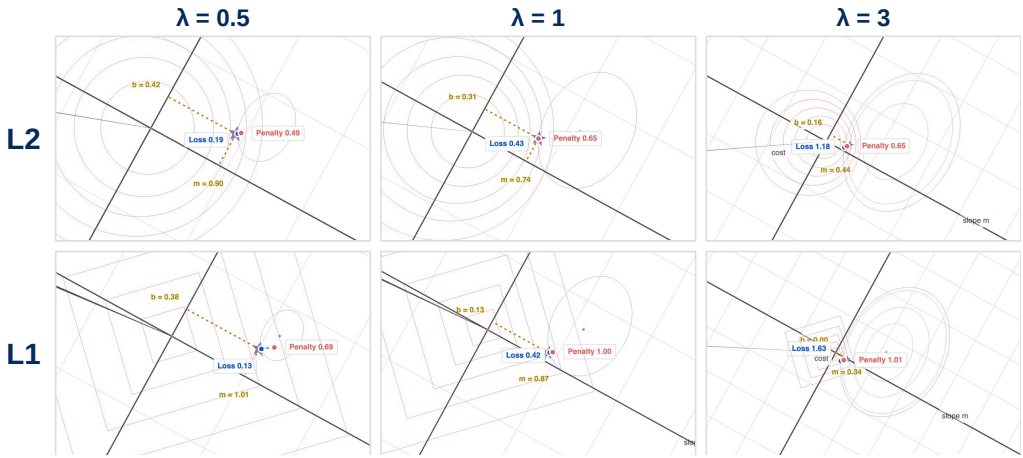
When we formalize **regularization** as a **cost**, we modify the **loss function** such that increasing the weights incurs a "cost".



We use an argument called λ to determine how heavily the **cost** is weighed. A larger λ increases the **cost**.

Regularization as a cost, increasing λ

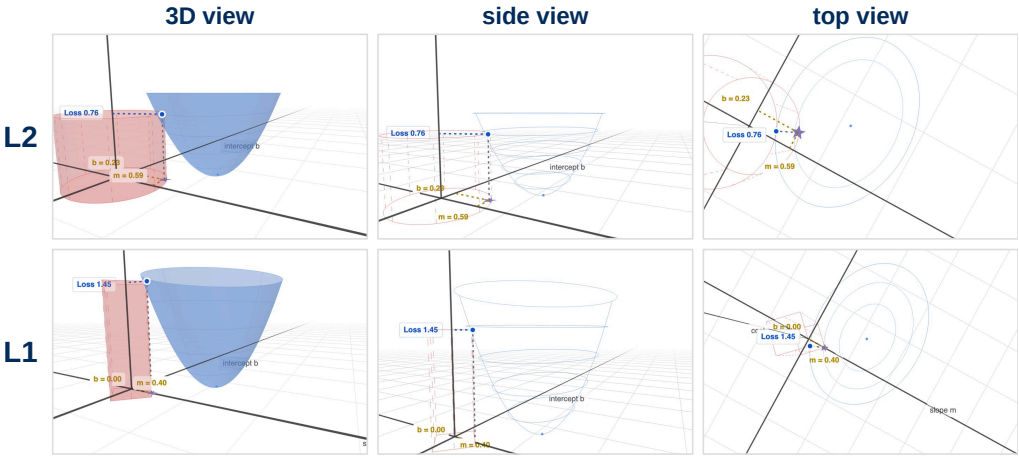
A larger λ increases the **cost** of our weights, typically resulting in the optimum being pulled closer to the origin.



Regularization as a budget

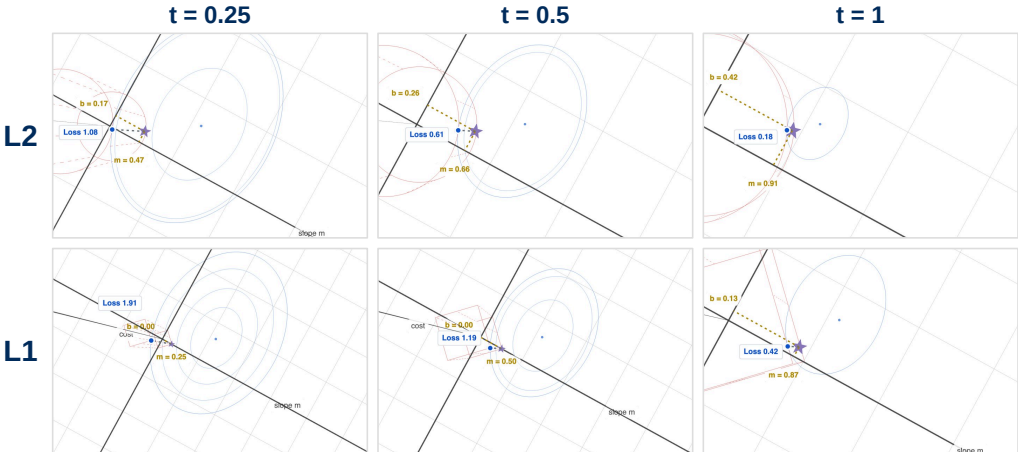
When we formalize **regularization** as a **budget**, we constrain our **loss function** such that the weights chosen must remain within "budget".

Regularization as a budget



Increasing t

A larger t increases the **budget** for our weights, typically resulting in the optimum being closer to the unconstrained optimum.



MAP estimation

L1 and **L2 regularization** are a form of **MAP Estimation**.

MAP stands for _____ which is a **Bayesian** approach for **applying a prior to our weights**.

Review

Complete the review: **L1 regularization** penalizes _____. **L2 regularization** penalizes _____. In a penalty formulation, increasing **lambda** makes the penalty _____. In a budget formulation, increasing the budget makes the constraint _____.

Matching: regression and regularization

Write the letter of the matching definition next to each term.

___ **Ordinary Least Squares**	**A.** Penalizes the squared size of the weights, encouraging small weights
___ **Maximum Likelihood Estimation**	**B.** Controls how heavily the regularization cost is weighed
___ **MAP Estimation**	**C.** The budget the weights must remain within
___ **L2 (Ridge)**	**D.** The model memorizes noise in the training data instead of the general pattern
___ **L1 (LASSO)**	**E.** The relationship between model complexity, training accuracy, and accuracy on unseen data
___ λ	**F.** Solving for the weights by setting the gradient of the loss equal to 0
___ **t**	**G.** Estimating the parameters that make the observed data most likely
___ **Underfitting**	**H.** The model is too weak to capture the underlying patterns in the data
___ **Overfitting**	**I.** Penalizes the absolute size of the weights, driving some to exactly 0
___ **Bias–variance tradeoff**	**J.** A Bayesian approach for applying a prior to our weights